25900236

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Core Course (DSC) - MG3DSCBIF200-SEQUENCE INFORMATICS
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest (1) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions. Answer any **5** questions
Each question carries **2** marks

1. State sanger sequencing method? [U]/[CO1]
    - Developed by Frederick Sanger in 1977. Used to determine the nucleotide sequence of DNA. Use chain termination (2 marks for 2 relevant points)
2. How does the choice of BLAST variant depend on whether the input sequence is nucleotide or protein? [U]/[CO2]
    - For each correct BLAST variant and input, 1 mark each.
3. Indicate why is protein family identification important? [A]/[CO4]
    - Predict the function, evolutionary relationships, protein classification, pathway and ontology identification etc (1 mark for each relevant point)
4. How can multiple sequence alignment assist in the annotation of newly sequenced genomes? [U]/[CO2]
    - Identifying genes, regulatory elements, conserved regions, motifs, and functional domains shared with known genomes. Comparison with homologous sequences, protein families etc. help to predict gene function. (1 mark each for a relevant point)
5. Give two examples of widely used tools for pairwise alignment. [K]/[CO2, CO3]
    - Any two from BLAST, FASTA, EMBOSS Needle with one line description (1 mark for each)
6. Define main purpose of a primer in sequencing? [U]/[CO1]

- A primer provides a starting site for DNA polymerase to begin synthesis during sequencing. It ensures specific binding to the target region, allowing accurate reading of the DNA sequence. (2 points 2 marks)

7. Analyze why PSI-BLAST is considered more powerful for protein family detection compared to PHI-BLAST. [An]/[CO3]

- PSI-BLAST allow detect distant homologs within protein families. Helps in identifying conserved patterns and motifs shared by protein families. (1 mark for each point)

[2x5 = 10]

Part B

Short Essay Questions. Answer any **4** questions
Each question carries **5** marks

8. Identify one biological application where the Needleman-Wunsch algorithm is more suitable than local alignment methods. [U]/[CO2]

- Define Needleman-Wunsch algorithm and its purpose (2 marks).
- Explain any application from phylogenetic analysis, functional annotation, study orthologs, analyze mutations etc (3 marks)

9. Discuss how dot plots help in identifying repeated sequences and palindromic regions. [U]/[CO2, CO3]

- What are dot plots (1 mark)
- Extra diagonals indicate repeated or duplicated regions (2 marks)
- Palindromic regions show up as diagonals perpendicular. (2 marks)

10. Define the tools used for MSA. [U]/[CO2]

- Mention any three tools from CLUSTAL Omega, ClustalW, ClustalX MUSCLE, T-Coffee (1 marks)
- Features of any two tools (3 marks)
- Application (1 mark)

11. Describe how a scoring matrix influences alignment results? [U]/[CO2]

- Define scoring matrices and name them (2 mark)
- Explain how different matrices scores matches, mismatches and gap penalties differently. (3 marks)

12. Compare sequence profile with regular expression in detecting motif. [An]/[CO3]

- Define sequence profile – 1 mark
- Define regular expression – 1 mark
- Compare differences in information provided, sensitivity, representation, use case etc. (any 2 points compared- 3 marks)

13. Explain the advantage of profile in studying conserved regions. [U]/[CO2]

- Define profiles and conserved regions- 2 mark
- Identifying functionally or structurally important regions, find remote homologs, high sensitivity, protein family assignment etc. (explain any two points - 3 marks)

[5x4 = 20]

Part C

Long Essay Questions. Answer any **2** questions
Each question carries **10** marks

14. Comprehend the significance of "Blocks" in BLOSUM matrices. [U]/[CO2]

- Defining scoring matrices (1 marks)
- SCRs and BLOCKS (2 marks)
- How BLOSUM matrices are created (5 marks)
- Significance of BLOSUM matrices (2 marks)

15. Explain how multiple sequence alignment is used in evolutionary studies and species identification. [A]/[CO4]

- Define MSA (2 marks)
- Role of MSA in identifying conserved and variable regions, DNA barcoding, detecting species-specific patterns etc (4 marks)
- Role of MSA in phylogenetic reconstruction and major tools (4 marks).

16. Explain the principle and steps of Maxam-Gilbert sequencing. [U]/[CO1]

- Brief description – 2 marks
- Principle. (2 marks)
- Steps of Maxam-Gilbert sequencing (4 marks)
- Diagram- 2 marks

17. Describe the importance of functional annotation in genome projects. [A]/[CO5]

- Define Functional annotation (2 marks)
- Explain role in Identification of Gene Function, Understanding Metabolic and Signaling Pathways, Comparative Genomics and Evolutionary Insights, Disease and Drug Target Identification, etc. (8 marks)

[10x2=20]